

Kushagra Sinha

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Technical Skills

Languages: Python, Go, Java, SQL, JavaScript, C++

Backend & Distributed Systems: Microservices, REST APIs, gRPC, Protobuf, Event-Driven Architecture, Apache Kafka, RabbitMQ, Concurrency, Multithreading, FastAPI, Flask

Databases & Storage: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, Cassandra, ClickHouse

Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, SNS, CloudWatch), Docker, Kubernetes, Terraform, Linux

CI/CD & Observability: Jenkins, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Kibana

AI / LLM Systems: LLM Agents, Multi-Agent Systems, Agent Orchestration, RAG, Vector Search, LLM Evaluation, LangGraph, LangChain, LangSmith, FAISS, Pinecone

Work Experience

Software Engineer Intern, Tesla – Austin, TX

Aug 2025 – Dec 2025

- Built a **natural language dashboard configuration platform** using Python, FastAPI, Elasticsearch, and LangGraph that translated natural language queries into validated Elasticsearch DSL through deterministic templates and a human-in-the-loop confirmation step, reducing dashboard widget creation time by **90%**
- Developed a **dataset onboarding service** that transformed Excel based specifications into validated JSON schemas using FastAPI, Pydantic, LLM and concurrent processing, reducing onboarding time from **several days to under an hour**
- Engineered a **conversational search system** using Python, Elasticsearch, Redis, and LangGraph that performed retrieval across assets, cases, and ticket datasets while maintaining session context across interactions, improving retrieval accuracy by **80%**
- Built a **retrieval augmented email generation API** using Python, ClickHouse vector search, and embeddings to automatically draft context aware emails from historical cases, user data, and internal knowledge sources, reducing draft creation time and ensuring consistent, relevant content
- Designed an **idempotent ticket creation agent** integrated with Microsoft Teams that verified ticket creation through backend API confirmation before responding to users, preventing duplicate ticket creation and eliminating hallucinated confirmations

Software Engineer, Qualcomm – Bangalore, India

Feb 2023 – Aug 2024

- Built a **multithreaded XML ETL pipeline** using Python, NumPy, MongoDB, and Elasticsearch to process hundreds of 8 to 9 GB GPU capture files, reducing processing time from **weeks to four days** and enabling the graphics driver team to query capture data
- Replaced a FIFO CI/CD workflow with a **RabbitMQ based priority scheduler** integrated with Jenkins that automatically promoted urgent release builds, eliminating manual queue intervention by engineering teams

Software Engineer, Zomato – Gurugram, India

Feb 2022 – Dec 2022

- Built an **event driven inventory alerting service** using Go, Kafka, AWS SNS, Redis, and MySQL that consumed demand forecasts and generated automated restocking alerts for kitchen administrators, reducing stock outs across cloud kitchens
- Owned **backend payment integrations** for Zomato Gold using Go, DynamoDB, gRPC, and Protobuf, implementing retries, circuit breakers, and idempotent writes to support more than **100,000 daily transactions** under production load, monitored with AWS CloudWatch
- Developed **backend APIs for the Zomato Instant Admin platform** secured with JWT authentication and role based access control, enabling sales teams to onboard cloud kitchens, manage menus, and update pricing without engineer operated SQL workflows, backed by AWS RDS and S3

Software Engineer, BlueStone – Bangalore, India

Feb 2021 – Dec 2021

- Engineered **catalog and pricing pipelines** using Java and PostgreSQL (AWS RDS) with transactional rollback and idempotent batch processing across **10,000+ SKUs**, preventing partial catalog updates and ensuring data consistency

Academic Projects

RolloutX – Distributed Feature Flag Platform

github.com/sinhakushagra21/RolloutX

- Built a **distributed feature flag platform** with separate control and data planes, where a transactional outbox publishes immutable versioned snapshots to Kafka and stateless data-plane replicas evaluate flags in-process over lock-free snapshots at **211 ns per evaluation** (~4.7M evaluations/sec per core)
- Implemented deterministic, sticky percentage rollouts with MurmurHash3 bucketing, property-tested to hold the distribution within **±1% across 1,000,000 keys**, with sub-second kill switches and a last-known-good SDK fallback so flags keep evaluating during outages

FitGen – Stateful Multi-Agent Fitness Planner

github.com/sinhakushagra21/fitgenai

- Built a **stateful multi-agent fitness AI coach** with an LLM router that directs each turn to specialist diet and workout tools, vector search RAG over saved plans, and a plan evaluator with a refinement loop plus scope guardrails, backed by **220+ tests**

Education

Northeastern University – M.S., Software Engineering Systems

Aug 2024 - Jul 2026

PES University – B.Tech, Electronics & Communication Engineering

Aug 2016 - Aug 2020